Silicon Sampling on CES 2025: Where We Stand

Two branches, one benchmark · prompting vs fine-tuning · what holds up

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In brief. Both branches try to reproduce how 15 Canadian subgroups (5 regions × 3 age groups) answered 10 CES 2025 questions, using only data from before 2025. Fine-tuning works much better than prompting (18 vs 30 answers out of 100 in the wrong place). But **a four-year-old poll, CES 2021 reused unchanged, still beats both (15 out of 100)**. Both branches find the same reason: models get the direction of subgroup differences right but shrink them to about a third of their real size.

1. The two branches, in plain terms

The common test. We take 10 questions from the 2025 Canadian Election Study (CES 2025) and 15 groups of Canadians (5 regions × 3 age groups). For each group and question, we compare how the model's fake respondents answered with how real Canadians answered (N = 20,180). Nothing from CES 2025 is shown to the models: it serves only as the answer key.

The score: total variation (TV). The share of answers that would have to move to another option to match reality. TV = 0.18 means 18 answers out of 100 are in the wrong place. Lower is better.

Both branches follow the same three steps. They differ in what happens at each one.

	Justin: prompting	Hubert: fine-tuning
Step 1. Teach the model before any test	**No teaching.** An off-the-shelf model (Qwen 3.5, 9 billion parameters), used as is.	**The model studies real survey answers.** Llama 3.3 (70 billion parameters) is shown 50,000 real answers from 8 Canadian surveys (CES 2019 and 2021, Democracy Checkup 2019 to 2024). Each example is one real person: their profile and 7 of their answers, then their answer to one more question, which the model learns to predict.
Step 2. Create fake respondents who answers the test	**1,500 real people from CES 2021** (100 per group, picked at random). Each fake respondent is a copy of one of them, as they were in 2021.	**15,000 real people from Democracy Checkup 2024** (1,000 per group, picked to match the population). Each fake respondent is a copy of one of them, as they were in 2024.
Step 3. Ask the questions what the model reads	A short description of the person: region, age, gender, province, education, income, language, religion, born in Canada, **party they identify with, left–right position, 2019 vote**. Then all 10 questions at once.	A short description of the person: age, gender, education, province, income, language. Then **their own answers to 7 questions**: left–right, party, rich–poor gap, bilingualism, trust, ties with the US, equal rights. Then one CES 2025 question at a time.
Scale	1,500 requests (10 answers each)	150,000 requests (1 answer each)
Temperature	1.0 for both: the model's natural randomness, neither reduced nor amplified.	

What one request looks like (fine-tuning branch, shortened).

```
You are a respondent to an opinion survey conducted in 2025.
Age: 18 to 24 · Gender: Man · Province: British Columbia · Income: $100,000 and over …
Your answers to other questions in the survey:
- Left-right, 0 to 10 → 9
- In federal politics, you usually think of yourself as a → Conservative
- … (5 more)
Question: How much should the federal government spend on affordable housing?
Options: Spend less / Spend about the same as now / Spend more / Don't know
```

Justin's request is similar, but the person's details are a bulleted list (including party and 2019 vote) and the 10 questions come together.

Two differences to keep in mind. The fine-tuned model has more to go on: a person's real answers to related questions, and a 2024 snapshot instead of 2021. Its better score therefore mixes two things, the training and the richer description. And with 10 times more fake respondents per group, its scores carry less random noise.

Dropped along the way (tested, not worth keeping):

- *Bigger model* (Qwen 2.4T): worse than the 9B in every condition; it piles onto neutral answers.
- *Shorter descriptions* (demographics only, without party or left–right): party information is the largest single gain for prompting.
- *Changing the 7 questions per question* (picking those closest in meaning to the one asked): worse than always using the same 7, at every training size.
- *More training examples* (100,000 instead of 50,000): no better.
- *Stretching the model's group differences* (Justin's IRT correction, ×3): gains 0.01 overall and hurts 4 of 10 questions.
- *Model's national level + CES 2021 group differences*: 0.165, better than the model alone but worse than CES 2021 alone, and it needs a past poll on the same question. Even fed the real 2025 national figures, it does worse (0.098) than CES 2021 gaps (0.064).

2. Scoreboard

Mean TV over the 150 subgroups, and over the 10 national distributions. Green is closer to reality, red further. Grey rows use CES 2025 itself and only show what is reachable.

Source	Branch	By subgroup	National
Human floor (two halves of CES 2025)	Reference	0.062	0.062
Real 2025 national for every subgroup	Reference	0.085	0
CES 2021, same subgroup, unchanged	Past poll	0.151	0.141
CES 2021 regression (same descriptions as Justin)	Past poll + statistics	0.175	0.150
Fine-tuned Llama 3.3 70B (50k examples)	Hubert	0.178	0.157
Prompted Qwen 3.5 9B (full description)	Justin	0.302	0.258
Llama 3.3 70B before teaching (step 1)	Hubert	0.325	0.296

- **Fine-tuning roughly halves the error** of the same model before teaching (0.325 → 0.178) and does much better than the best prompting (0.302).
- **A fine-tuned model is about as good as a standard regression fitted on 2021** (0.178 vs 0.175 by subgroup), but neither beats simply reusing CES 2021.

What these scores are worth, in real respondents. A real poll also misses, by chance, and less so the more people it asks. We simulated real polls of different sizes drawn from CES 2025 and measured their TV the same way. The model is "worth" the poll size that misses by as much.

Mean TV by subgroup	Same error as a real poll of… per subgroup	…in total (15 subgroups)
0.178 (fine-tuned model today)	about 14 people	about 200
0.151 (CES 2021 reused)	about 20	about 300
0.10 (target: useful)	**about 40**	**about 600**
0.08 (target: reliable)	**about 65**	**about 1,000, a standard poll**

One difference: a real poll's error is random and shrinks as it grows; a model's error is a bias and does not shrink with more fake respondents.

3. By question

Mean TV over the 15 subgroups. **Bold** marks the best of the three forecasts on each question.

Question	CES 2021 unchanged	Fine-tuned Llama 70B	Prompted Qwen 9B	Floor
Spending: affordable housing	**0.08**	0.10	0.13	0.05
Satisfaction with democracy	**0.06**	0.12	0.23	0.05
Medical assistance in dying	**0.09**	0.14	0.23	0.09
Jobs before environment	**0.13**	0.15	0.31	0.10
Spending: defence	0.31	**0.17**	0.24	0.04
Pipelines	**0.16**	0.17	0.31	0.10
Spending: national childcare	**0.09**	0.18	0.31	0.06
Personal finances, past year	**0.13**	0.21	0.21	0.04
Spending: reconciliation	**0.17**	0.24	0.44	0.05
Immigration levels	0.31	**0.30**	0.60	0.04
Mean	**0.151**	0.178	0.302	0.062

- **CES 2021 wins on 8 of 10 questions.** The fine-tuned model wins on **defence**, where opinion moved sharply after 2021 and its training covers surveys up to 2024. On **immigration** both are poor.
- **The fine-tuned model beats prompting on every question**, by the widest margin on immigration (0.30 vs 0.60) and reconciliation (0.24 vs 0.44).
- **Shared misses.** Immigration: 62% of Canadians want fewer immigrants; prompted models chose it 2–4% of the time, and Azure's content filter removed most immigration items from fine-tuning. Reconciliation: prompted models say "spend more" 68–91% of the time, against 22% of Canadians.

4. What both branches find

- **Subgroups are flattened.** Hubert measures model subgroup gaps at about 31% of their real size (correlation 0.42; CES 2021: 69%, 0.71). Justin finds the same with an item response model: right direction on every question, slope at roughly a third of the human one.
- **Most of the error is the national level.** For fine-tuning and CES 2021 alike, the national error (0.157, 0.141) is close to the subgroup error. Giving CES 2021 gaps the real 2025 national figure drops the error to 0.064, almost the human floor (0.062).
- **On repeated questions, a past poll is the bar, and it is high.** All 10 questions were asked in CES 2021. Silicon sampling should add value on questions with no past version.

5. Next steps

A. Change the test: questions never asked before

A client will ask questions that no past poll has asked. Our 10 questions were all in CES 2021, which is why reusing CES 2021 is such a strong rival here: in real use, it would not exist.

- **Add a second set of questions:** CES 2025 items never asked in earlier surveys, and removed from all training data. This is the score that matters for the product.
- **Keep the current set** with CES 2021 as the target to reach (0.151): on repeated questions, a model must at least match a past poll.

B. Make the fine-tuned model know each person better

The models flatten subgroups because they know too little about each person and fall back on the average. Today they see 7 of the person's answers.

1. **Show 15 to 20 answers instead of 7:** a fixed core of themes, plus 5 or so answers picked for their closeness to the question asked. Cost is not a constraint.
2. **Start at 50,000 examples**, then try 100,000 and 150,000 with the best recipe, training each size twice to see how much results vary from one training run to the next.

Temperature stays at 1.0. Raising it makes every group answer more at random, which flattens them further; lowering it pushes everyone toward the most common answer.

C. A new approach: from a dossier to a distribution

> **Today**, the model plays 15,000 fake respondents, one at a time, and we count their answers. **Instead**, the model would act like an analyst: we hand it a **dossier** on the new question and it writes down, in one go, **the percentage of each answer in each of the 15 subgroups**.

What the dossier contains (all from past surveys, nothing about the new question's answers):

- the question's wording and options;
- how Canadians answered **similar past questions**, nationally;
- how **each subgroup differed from the national figure** on those questions (for instance, how far Quebec 18–34 sits from the national figure on each spending item);
- simple summaries of the main attitude dimensions (left–right, trust, economy) by subgroup.

What it returns: a table of 15 subgroups × the answer options, in percentages.

Why it is promising.

- It gives exactly what a client wants, for a question nobody asked yet.
- It uses what we know is stable: subgroups differ in similar ways across related questions.
- One request per question instead of 15,000.
- A published study finds language models **describe** a group's opinion distribution more accurately than they **reproduce** it by role-playing respondents one by one (Meister, Guestrin and Hashimoto, NAACL 2025, *Benchmarking Distributional Alignment of Large Language Models*).

How we would train it. Every question in our 8 surveys becomes one practice exercise: build its dossier without that question, ask the model for the 15 distributions, compare with the real answers. That gives thousands of exercises. Two ways to learn from them:

1. **By example** (as today): the model is shown the right table and learns to write it.
2. **By reward**: the model proposes a table, a small program scores it (the closer to reality, the higher), and the model learns to raise its score. This targets our real goal directly. Azure offers this for some OpenAI models

(o4-mini; GPT-5 on invitation), with a scoring program we write ourselves, but not for Llama according to its documentation.

Practice and test questions must cover different topics, or the model would see near-copies of the test questions while practising.

How we would judge it. Against the fine-tuned model of section B, and against a plain statistical version of the same idea with no language model (predict a new question's subgroup gaps from similar questions; Justin's August "item encoder"). If the language model cannot beat that, it adds nothing.

First step, no training needed: build the dossier for our 10 questions as if they were new (leaving them out of it) and ask an off-the-shelf model. About ten requests.

D. Already measured, for later

- **A product option:** one real national figure for the new question (a small national poll) plus the model's subgroup differences gives **0.082**, worth about 60 real respondents per subgroup. The model then only does what it does well: telling how groups differ.
- **For the next note's appendix:** does the model use the survey year? Same person, same question, with "2019" swapped for "2025" in the prompt. If the national answer barely moves, the model answers for an average of 2019–2024.

Comparability: Justin's subgroup scores use 100 synthetic respondents per subgroup (Hubert: 1,000), which adds a few points of sampling noise; his scores are weighted by population (Hubert: equal weights; 0.174 vs 0.178 for the fine-tuned model). Logit and prompting figures as reported in Justin's notes of October 12–14.